IMDb Movie Review Sentiment Analysis

1. Introduction

Sentiment Analysis is a Natural Language Processing (NLP) technique used to determine whether a piece of text expresses a positive, negative, or neutral sentiment.

This project analyzes IMDb movie reviews and classifies them as Positive or Negative using machine learning and NLP techniques.

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2. Objective

The primary objective of this project is to build a sentiment classification model that can automatically identify the sentiment of movie reviews.

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3. Problem Statement

With thousands of online reviews generated daily, manually analyzing customer opinions becomes difficult and time-consuming.

The goal of this project is to automate sentiment detection using NLP and machine learning algorithms.

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4. Dataset Description

The dataset contains IMDb movie reviews with sentiment labels.

Column	Description
Review	Movie review text
Sentiment	Positive or Negative

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5. Data Preprocessing

The following preprocessing steps were performed:

Lowercase Conversion

Converted all text to lowercase.

HTML Tag Removal

Removed unnecessary HTML tags from reviews.

Special Character Removal

Removed punctuation and special symbols.

Stopword Removal

Removed common words such as "the", "is", and "and".

Lemmatization

Converted words to their root form.

Examples:

- Running → Run
- Dogs → Dog
- Better → Good

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6. Exploratory Data Analysis

EDA was performed to understand:

- Dataset structure
- Missing values
- Duplicate records
- Sentiment distribution
- Review length patterns

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7. Feature Engineering

TF-IDF Vectorization was used to convert text reviews into numerical features suitable for machine learning models.

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8. Model Development

The dataset was split into training and testing sets.

The machine learning model was trained on processed review data and evaluated using classification metrics.

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9. Evaluation Metrics

The model performance was measured using:

- Accuracy Score
- Confusion Matrix
- Classification Report
- Precision
- Recall
- F1-Score

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10. Results

The sentiment analysis model successfully classified movie reviews into Positive and Negative categories.

The preprocessing pipeline significantly improved text quality and model performance.

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11. Business Applications

- Customer Feedback Analysis
- Product Review Monitoring
- Social Media Sentiment Analysis
- Brand Reputation Management
- Market Research

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12. Conclusion

This project demonstrates how NLP and Machine Learning can be used to analyze textual data and automatically determine sentiment. It highlights the importance of text preprocessing, feature engineering, and model evaluation in building effective sentiment analysis systems.

Author

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Skills Demonstrated: Python | NLP | Machine Learning | Data Cleaning | TF-IDF | Sentiment Analysis | Data Visualization